

Mathematics Playbook

All units in one book: Numbers · Ratio · Fractions · Algebra · Geometry · Mensuration · Data

ICSE Class 6 · Full Book · Final Exam Ready

Play & Learn Notes · Independent study material, not affiliated with CISCE.
Chapter names and order can differ slightly between schools and publishers. Follow your school's exam syllabus.

How to use this playbook

For the teacher: each chapter has short notes, a 🧠 memory trick, a 🎲 classroom game (5–10 minutes) and a ✏️ quick check. The answers are at the end of the book.

For the student: read the notes, say the memory trick out loud, then try the quick check without looking.

Unit	Chapters covered
1. Number System	Knowing our numbers, Indian & International system, place value, estimation, whole numbers, integers, number line, divisibility, factors & multiples, HCF & LCM, sets
2. Ratio & Proportion	Ratio, proportion, unitary method
3. Fractions & Decimals	Fractions, decimal fractions, percent, speed–distance–time
4. Algebra	Basic terms, operations, substitution, framing expressions, simple linear equations
5. Geometry	Basic concepts, angles, lines, triangles, quadrilaterals, polygons, circle, symmetry, solids
6. Mensuration	Perimeter and area
7. Data Handling	Data, pictographs, bar graphs, mean and median

Final exam countdown plan (suggested)

Week	Revise
1–2	Number System and Sets
3	Ratio, Proportion, Unitary Method
4–5	Fractions, Decimals, Percent, Speed
6–7	Algebra and Linear Equations
8–9	Geometry
10	Mensuration and Data Handling
11–12	Mixed revision tests and the full-book quick checks

1. Knowing Our Numbers

Indian and International systems

Indian	Digits	International
Thousand (1,000)	4	Thousand (1,000)
Lakh (1,00,000)	6	Hundred thousand (100,000)
Ten lakh (10,00,000)	7	Million (1,000,000)
Crore (1,00,00,000)	8	Ten million (10,000,000)
Ten crore	9	Hundred million
Arab (100 crore)	10	Billion

- **Indian commas:** the first comma comes after 3 digits from the right, then after every 2 digits: **5,23,47,816**
- **International commas:** a comma after every 3 digits: **52,347,816**

> 🧠 **Memory Trick**
> **Indian = 3-2-2-2** (like the Mumbai local: first a big compartment, then small ones). **International = 3-3-3** (like an escalator, all steps are equal). **10 lakh = 1 million**, and **1 crore = 10 million**.

Place value and face value

- **Face value** = the digit itself. **Place value** = digit × its place.
- In 4,**7**3,215 the face value of 7 is **7**, and its place value is **70,000**.

Estimation (rounding off)

- Look at the digit just to the right of the place you are rounding to. If it is **5 or more, round up**. If it is **less than 5, round down**.
- 4,672 to the nearest hundred → look at 7 → round up → **4,700**
- 83,419 to the nearest thousand → look at 4 → round down → **83,000**

> 🎲 **Play It in Class: Number Antakshari**
> One student says a big number, like "6,45,302". The next student must say it in words, then give a new number. If they say it wrong, they sit down. Play one round in the Indian system and one in the International system.

Roman numerals

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

- A smaller numeral **before** a bigger one means subtract: IV = 4, IX = 9, XL = 40, XC = 90
- A smaller numeral **after** a bigger one means add: VI = 6, LX = 60

- I, X, C and M can repeat up to 3 times. **V, L and D never repeat.**

> 🧠 **Memory Trick**
> "**I** **V**alue **X**-tra **L**arge **C**old **D**rinks **M**ore": that's the order I, V, X, L, C, D, M.

> ✏️ **Quick Check**
> 1. Write 7,08,45,000 in words (Indian system).
> 2. Insert commas in 4567890 using the International system.
> 3. Round off 56,749 to the nearest thousand.
> 4. Write 94 in Roman numerals.

2. Whole Numbers, Integers and the Number Line

Kinds of numbers

- **Natural numbers (N):** 1, 2, 3, … (counting numbers, with no 0)
- **Whole numbers (W):** 0, 1, 2, 3, … (natural numbers + 0)
- **Integers (Z):** …, −3, −2, −1, 0, 1, 2, 3, …
- The **successor** of a number is that number + 1, and its **predecessor** is that number − 1. 0 has no predecessor among whole numbers.

Properties (whole numbers)

Property	Addition	Multiplication
Closure	Always a whole number ✔️	✔️
Commutative	a + b = b + a	a × b = b × a
Associative	(a + b) + c = a + (b + c)	(a × b) × c = a × (b × c)
Identity	a + 0 = a	a × 1 = a
Distributive	a × (b + c) = a × b + a × c	

- Subtraction and division are **not** commutative: 5 − 3 ≠ 3 − 5.
- Division by 0 is **not defined**.

Integers on the number line

- Numbers to the right are bigger. **−2 > −7**, because −2 is closer to the right.
- **Adding** a positive number means moving right. **Adding** a negative number means moving left.
- **Subtracting** an integer means adding its opposite: 5 − (−3) = 5 + 3 = **8**

Multiply or divide	Result
(+) × (+)	+
(−) × (−)	+
(+) × (−) or (−) × (+)	−

🧠 **Memory Trick**

"A friend of a friend is a friend. An enemy of an enemy is a friend." (+)(+) = +, (−)(−) = +, but a friend with an enemy (+)(−) = −

📍 **Mumbai Connection: Temperature and lift buttons**

A Mumbai building has floors B2, B1 (basements), G, 1, 2, 3 … That's an integer line! B2 = −2, G = 0. Going from B2 up to the 3rd floor = 3 − (−2) = **5 floors**.

🎲 Play It in Class: Human Number Line

Stick cards from −10 to +10 on the floor. Give a student a sum like "−3 + 5". They stand on −3 and walk 5 steps right, landing on +2. The class checks the answer.

✏️ Quick Check

1. Write the predecessor of 1,00,000.
2. Find: (−12) + 7 − (−5)
3. Find: (−6) × (−4) and (−36) ÷ 9
4. Arrange in ascending order: 3, −8, 0, −1, 5

3. Divisibility, Factors, HCF and LCM

Divisibility rules

Divisible by	Rule	Example
2	The last digit is 0, 2, 4, 6 or 8	538
3	The sum of the digits is divisible by 3	471 → 4 + 7 + 1 = 12
4	The last 2 digits are divisible by 4	3,716 → 16
5	The last digit is 0 or 5	945
6	Divisible by both 2 and 3	312
8	The last 3 digits are divisible by 8	5,120 → 120
9	The sum of the digits is divisible by 9	729 → 18
10	The last digit is 0	670
11	(sum of digits in odd places) − (sum in even places) = 0 or a multiple of 11	2,728 → (8 + 7) − (2 + 2) = 11

Factors and multiples

- A **factor** divides a number exactly. A **multiple** is the number × 1, 2, 3, ...
- **Prime:** exactly 2 factors (2, 3, 5, 7, 11, ...). **Composite:** more than 2 factors.
- **1 is neither prime nor composite. 2 is the only even prime.**
- **Co-primes:** numbers whose HCF is 1 (8 and 15). **Twin primes:** primes that differ by 2 (11 and 13).
- **Prime factorisation:** 60 = 2 × 2 × 3 × 5

HCF and LCM

- **HCF** (Highest Common Factor) is the biggest number that divides all of them.
- **LCM** (Lowest Common Multiple) is the smallest number that all of them divide into.
- **The golden rule (for 2 numbers):** HCF × LCM = product of the numbers

Example: 12 = 2 × 2 × 3, 18 = 2 × 3 × 3 HCF = 2 × 3 = **6** (common factors, lowest powers). LCM = 2 × 2 × 3 × 3 = **36** (all factors, highest powers). Check: 6 × 36 = 216 = 12 × 18 ✔️

> 🧠 **Memory Trick**
> **HCF = "share"** (it's about dividing things into equal groups, like ribbons cut into equal pieces). **LCM = "meet again"** (it's about things that repeat, like two bells or two buses meeting again).

> ⭐ **Exam Tip**
> Word problems: if the question says "**largest** size / **maximum** number of equal pieces", use **HCF**. If it says "**together again** / **least** number / **at the same time**", use **LCM**.

📍 Mumbai Connection

Two Mumbai local trains leave Churchgate together. One returns every 12 minutes and the other every 18 minutes. When will they leave together again? LCM(12, 18) = **36 minutes**.

🎲 Play It in Class: Fizz-Buzz Primes

Count around the class. On a multiple of 3 say "Fizz", on a multiple of 5 say "Buzz", and on both say "Fizz-Buzz". Round 2: clap on every prime number.

✏️ Quick Check

1. Is 5,148 divisible by 4? By 9? By 11?
2. Write the prime factorisation of 84.
3. Find the HCF and LCM of 24 and 36.
4. The HCF of two numbers is 4 and their LCM is 48. One number is 16. Find the other.

4. Sets

- A **set** is a well-defined collection of objects. "Good students" is **not** well-defined. "Vowels in English" **is** well-defined.
- Elements are written in curly brackets: A = {a, e, i, o, u}
- ∈ means "belongs to" and ∉ means "does not belong to": e ∈ A, b ∉ A
- **Roster form:** {2, 4, 6, 8}. **Set-builder form:** {x : x is an even number, x < 10}
- **Cardinal number n(A)** is the number of elements: n({a, e, i, o, u}) = 5

Type	Meaning	Example
Empty (null) set ⌀ or { }	No elements	Months with 32 days
Singleton	Exactly 1 element	{0}
Finite	Elements can be counted	Days of the week
Infinite	Elements cannot be counted to the end	Natural numbers
Equal sets	Exactly the same elements	{1, 2, 3} = {3, 1, 2}
Equivalent sets	The same number of elements	{a, b} and {5, 9}
Subset ⊆	Every element of A is in B	{1, 2} ⊆ {1, 2, 3}

- Repeated elements are written **once**: the letters of "MUMBAI" = {M, U, B, A, I}, so n = 5

🧠 **Memory Trick**

Equal = same members. Equivalent = same count. ("Equal twins, equivalent teams") **{0} is NOT empty.** It has one element, 0.

🎲 **Play It in Class: Set Sorting Race**

Write 20 words on the board (mango, Pune, 7, tiger, Nagpur, 13…). Teams race to sort them into sets: {fruits}, {cities in Maharashtra}, {prime numbers}, {animals}. Then each team writes one set in roster form and set-builder form.

✏️ **Quick Check**

1. Write the set of letters of the word "ELEPHANT" in roster form, and find n(A).
2. Is {0} an empty set? Why?
3. Write in set-builder form: {1, 4, 9, 16, 25}
4. Are A = {p, q, r} and B = {1, 2, 3} equal or equivalent?

5. Ratio, Proportion and Unitary Method

Ratio

- A **ratio** compares two quantities **of the same kind and in the same unit**: a : b = a/b
- Always simplify: 24 : 36 = **2 : 3** (divide by the HCF, 12)
- **Convert to the same unit first:** 50 paise : ₹2 = 50 : 200 = **1 : 4**
- A ratio has **no unit**.
- **Dividing in a ratio:** share ₹600 between A and B in the ratio 2 : 3. Total parts = 5, so A = 2/5 × 600 = **₹240** and B = **₹360**

Proportion

- Four numbers are in proportion if a : b = c : d, written **a : b :: c : d**
- **Product of extremes = product of means:** a × d = b × c
- Are 3, 5, 12, 20 in proportion? 3 × 20 = 60 and 5 × 12 = 60 ✔️
- **Continued proportion:** a : b = b : c, so b^2 = a × c (b is the **mean proportional**)

Unitary method

- Find the value of **ONE** first, then of **many**.
- 6 pens cost ₹90. So 1 pen = 90 ÷ 6 = ₹15, and 10 pens = 15 × 10 = **₹150**

🧠 **Memory Trick**

"Extremes on the outside, means in the middle": a : **b :: c** : d. Outside × outside = inside × inside. **Unitary method = "One, then Many"** (like finding the price of 1 vada pav before buying for the whole class!)

📍 **Mumbai Connection**

A BEST bus pass costs ₹750 for 30 days. What is the cost per day? 750 ÷ 30 = **₹25 per day**.

🎲 **Play It in Class: Recipe Scaling**

"A recipe for 4 people needs 2 cups of rice. How much rice do we need for 10 people?" Teams scale a full recipe (pav bhaji!) up or down using the unitary method. The fastest correct team wins.

✏️ **Quick Check**

1. Simplify the ratio 45 minutes : 2 hours.
2. Divide ₹1,400 between Riya and Aarav in the ratio 3 : 4.
3. Are 4, 6, 10, 15 in proportion?
4. 8 notebooks cost ₹240. What is the cost of 13 notebooks?

6. Fractions

- **Proper:** numerator < denominator (3/7). **Improper:** numerator ≥ denominator (9/4). **Mixed:** 2 1/4
- **Converting:** 9/4 = 2 1/4 (9 ÷ 4 = 2, remainder 1), and 3 2/5 = (3 × 5 + 2)/5 = 17/5
- **Equivalent fractions:** multiply or divide the top and bottom by the same number: 2/3 = 4/6 = 8/12
- **Lowest terms:** divide by the HCF: 18/24 = **3/4**
- **Like fractions** have the same denominator. **Unlike fractions** have different denominators.

Comparing

- Like fractions: the bigger numerator is bigger.
- Unlike fractions: make the denominators the same using the LCM, or cross-multiply. For 3/5 vs 2/3: 3 × 3 = 9 and 5 × 2 = 10, so **3/5 < 2/3**

Operations

Operation	How	Example
Add or subtract	Make the denominators equal (LCM), then add or subtract the numerators	3/4 + 1/6 = 9/12 + 2/12 = 11/12
Multiply	top × top, bottom × bottom	2/3 × 9/4 = 18/12 = 3/2
Divide	Multiply by the reciprocal	3/4 ÷ 1/2 = 3/4 × 2/1 = 3/2
"of"	Means multiply	2/5 of 30 = 12

BODMAS (order of operations)

Brackets → **O**f → **D**ivision → **M**ultiplication → **A**ddition → **S**ubtraction Brackets order: ( ) first, then { }, then [ ]

🧠 **Memory Trick**

Dividing fractions: "Keep, Change, Flip." Keep the first fraction, change ÷ to ×, flip the second. **BODMAS: "Big Old Dogs Must Always Sleep."**

🎲 **Play It in Class: Pizza Fractions**

Draw 4 big circles (pizzas). Students colour slices to show 1/2, 2/4, 3/6 and 4/8, and discover they are all equal. Then "order pizzas" for the class: 3/4 + 1/2 of a pizza = how much?

✏️ **Quick Check**

1. Convert 29/6 into a mixed fraction.
2. Reduce 36/48 to its lowest terms.
3. Find: 2 1/3 + 1 3/4
4. Simplify: 1/2 + 3/4 × 2/3 (use BODMAS)

7. Decimals, Percent and Speed

Decimal fractions

- 0.7 = 7/10, 0.07 = 7/100, 0.007 = 7/1000
- **Place value:** in 23.456, 4 is tenths, 5 is hundredths and 6 is thousandths
- **Fraction to decimal:** divide (3/4 = 0.75), or make the denominator 10 or 100 (3/4 = 75/100 = 0.75)
- **Decimal to fraction:** 0.45 = 45/100 = **9/20**
- **Adding or subtracting:** line up the decimal points (fill gaps with 0s)
- **Multiplying:** multiply as normal, then count the total decimal places: 1.2 × 0.3 = 0.36
- **× 10, 100, 1000:** move the point right by 1, 2, 3 places. **÷ 10, 100, 1000:** move it left.

Percent

- **Percent = "out of 100"**: 25% = 25/100 = 1/4 = 0.25
- **Fraction to percent:** × 100 (3/5 × 100 = 60%)
- **Percent of a number:** 15% of 240 = 15/100 × 240 = **36**
- **What percent is a of b?** (a ÷ b) × 100. For 18 out of 24: 18/24 × 100 = **75%**

Fraction	1/2	1/4	3/4	1/5	1/10	1/3
Percent	50%	25%	75%	20%	10%	33 1/3%

Speed, distance and time

- **Speed = Distance ÷ Time. Distance = Speed × Time. Time = Distance ÷ Speed.**
- **km/h to m/s:** × 5/18. **m/s to km/h:** × 18/5. For example, 72 km/h = 72 × 5/18 = **20 m/s**

🧠 **Memory Trick**

The DST triangle: write **D** on top and **S** and **T** below it. Cover the one you want: D = S × T, S = D ÷ T, T = D ÷ S

📍 **Mumbai Connection**

The Mumbai–Pune distance is about 150 km. A car takes 3 hours. Speed = 150 ÷ 3 = **50 km/h**.

🎲 **Play It in Class: Shop Discount Game**

Make price tags for things like "School bag ₹800, 25% OFF". Students act as shoppers and work out the final price. The best saver wins.

✏️ Quick Check

1. Convert 0.125 into a fraction in its lowest terms.
2. Find: 4.6 + 0.38 + 12.05
3. A student scored 42 out of 60. Find the percentage.
4. Convert 54 km/h into m/s.

8. Algebra: Expressions and Substitution

Basic terms

- **Variable:** a letter that stands for a number (x, y, a). **Constant:** a fixed number (5, −3).
- **Term:** a part of an expression separated by + or −. In 3x + 2y − 7 there are **3 terms**.
- **Coefficient:** the number multiplied by the variable. In −5ab, the coefficient is **−5**.
- **Like terms:** the same variables with the same powers ($3x^2$ and $-7x^2$). **Unlike terms:** 3x and 3y.
- **Monomial** = 1 term, **binomial** = 2 terms, **trinomial** = 3 terms, **polynomial** = many terms
- **Degree:** the highest power of the variable. $4x^3 + 2x - 1$ has degree **3**.

Operations

- You can add or subtract **only like terms**: 5x + 3x − 2x = **6x**, and 3a + 4b + 2a = **5a + 4b**
- **Multiplying:** multiply the numbers, then add the powers: $3x^2 \times 4x^3 = \mathbf{12x^5}$
- **Dividing:** divide the numbers, then subtract the powers: $12a^5 \div 3a^2 = \mathbf{4a^3}$
- **Removing brackets:** a minus sign before a bracket changes every sign inside: −(2x − 3) = −2x + 3

Framing expressions

Words	Expression
5 more than x	x + 5
3 less than y	y − 3
Twice a number	2n
A number divided by 4	n/4
7 subtracted from 3 times p	3p − 7

Substitution

If a = 2 and b = −3: 3a + 2b = 3(2) + 2(−3) = 6 − 6 = **0**

🧠 **Memory Trick**

Like terms are like twins: same letters, same powers. Only twins can be added together!

🎲 **Play It in Class: Algebra Tiles Shopping**

Mango = m, banana = b. "Riya buys 3 mangoes and 2 bananas, and Aarav buys 1 mango and 4 bananas. Together?" (4m + 6b). Then: "A mango costs ₹20 and a banana costs ₹5. Find the total cost." (substitution: ₹110)

✏️ Quick Check

1. Write the coefficient of x in $-7xy$.
2. Simplify: $8a - 3b + 2a + 5b$
3. Write an expression: "4 less than twice x".
4. If $x = 3$ and $y = -2$, find $2x^2 - 3y$.

9. Simple Linear Equations

- An **equation** is a statement that two expressions are equal: 2x + 3 = 11
- **Linear** means the highest power of the variable is 1.
- **Solving:** do the **same** thing to both sides, or **transpose** (move a term across and change its sign).

Move across the = sign	It becomes
+	−
−	+
×	÷
÷	×

Example 1: 5x − 7 = 3x + 9 → 5x − 3x = 9 + 7 → 2x = 16 → **x = 8 Example 2:** x/3 + 2 = 6 → x/3 = 4 → **x = 12 Always check:** put the answer back in. For example 1: 5(8) − 7 = 33 and 3(8) + 9 = 33 ✔️

Word problems: four steps

1. Let the unknown be **x**.
2. Write the equation from the words.
3. Solve it.
4. Check the answer and write it in a sentence.

Example: "A number is doubled and 6 is added. The result is 30." 2x + 6 = 30 → 2x = 24 → **x = 12**

🧠 **Memory Trick**

"Cross the bridge, change your clothes!" When a term crosses the = bridge, its sign changes.

🎲 **Play It in Class: Mystery Number Detective**

The teacher thinks of a number: "I multiply it by 3, then subtract 4, and I get 11. What is my number?" Students write the equation (3x − 4 = 11) and solve it (x = 5). Then students take turns making their own riddles.

✏️ **Quick Check**

1. Solve: 4x + 5 = 29
2. Solve: 7x − 3 = 4x + 12
3. Solve: (x − 2)/5 = 3
4. The sum of a number and 15 is 42. Find the number.

10. Geometry: Lines and Angles

Basic terms

- **Point:** shows a position, has no size. **Line:** extends endlessly both ways. **Ray:** one end point, extends one way. **Line segment:** two end points, has a fixed length.
- **Collinear points** lie on the same line. **Intersecting lines** meet at one point. **Parallel lines** never meet (∥).
- **Concurrent lines:** three or more lines that pass through the same point.

Types of angles

Angle	Measure
Acute	Less than 90°
Right	Exactly 90°
Obtuse	Between 90° and 180°
Straight	Exactly 180°
Reflex	Between 180° and 360°
Complete	Exactly 360°

Pairs of angles

- **Complementary:** add up to **90°** (30° and 60°)
- **Supplementary:** add up to **180°** (110° and 70°)
- **Adjacent angles:** share a common arm and vertex
- **Linear pair:** adjacent angles on a straight line, which add up to **180°**
- **Vertically opposite angles** are **equal** (when two lines cross)
- **Parallel lines and a transversal:** corresponding angles are equal, alternate angles are equal, and co-interior angles add up to 180°

🧠 **Memory Trick**

C comes before S, and 90 comes before 180. Complementary = 90°, **S**upplementary = 180°. Or: **C**omplementary makes a **C**orner (90°). **S**upplementary makes a **S**traight line (180°).

🎲 **Play It in Class: Body Angles**

The teacher calls out "Acute!", "Right!", "Obtuse!", "Straight!" Students show the angle with their arms. Anyone who is wrong becomes the next caller.

✏️ Quick Check

1. Find the complement of 35°.
2. Find the supplement of 128°.
3. Two angles form a linear pair. One is 3 times the other. Find both.
4. Name the angle: 270°.

11. Triangles, Quadrilaterals, Polygons and Circles

Triangles

- **By sides:** equilateral (all 3 sides equal), isosceles (2 sides equal), scalene (no sides equal)
- **By angles:** acute-angled, right-angled, obtuse-angled
- **Angle sum property:** the three angles add up to **180°**
- **Exterior angle** = the sum of the two **interior opposite** angles
- The sum of any two sides is always **greater** than the third side.
- In an equilateral triangle, each angle = **60°**

Quadrilaterals (4 sides, angle sum = 360°)

Shape	Special property
Parallelogram	Opposite sides are parallel and equal, and opposite angles are equal
Rectangle	All angles are 90°, and the diagonals are equal
Square	All sides are equal, all angles are 90°, and the diagonals are equal and cross at 90°
Rhombus	All sides are equal, and the diagonals cross at 90°
Trapezium	Exactly one pair of parallel sides
Kite	Two pairs of equal adjacent sides

Polygons

- Triangle (3 sides), quadrilateral (4), pentagon (5), hexagon (6), heptagon (7), octagon (8), nonagon (9), decagon (10)
- **Angle sum of a polygon = (n − 2) × 180°.** For a hexagon: (6 − 2) × 180° = **720°**
- **Regular polygon:** all sides and all angles are equal

Circle

- **Centre**, **radius** (from the centre to the circle), **diameter** = 2 × radius (passes through the centre)
- **Chord:** joins two points on the circle. The diameter is the **longest chord**.
- **Arc:** a part of the circle. **Semicircle:** half a circle. **Sector:** a "pizza slice". **Segment:** the part between a chord and an arc.
- **Circumference:** the distance around the circle

> 🧠 **Memory Trick**
>
> **Polygon names: "Tri-Quad-Penta-Hexa-Hepta-Octa-Nona-Deca"**, said like a rap: 3, 4, 5, 6, 7, 8, 9, 10! **A square is a special rectangle AND a special rhombus.** It belongs to both families.

📍 Mumbai Connection: Shapes in our city

The Gateway of India's arch is a semicircle on top. The Bandra–Worli Sea Link cables form triangles. Ask students: "Find 5 shapes on your way to school."

🎲 Play It in Class: Shape Hunt

Teams get 5 minutes to find and list classroom objects for each shape: triangle, rectangle, square, circle, hexagon. Each correct object scores 1 point. A rare shape scores 2 points.

✏️ Quick Check

1. Two angles of a triangle are 65° and 45°. Find the third.
2. Find the angle sum of an octagon.
3. Three angles of a quadrilateral are 80°, 95° and 110°. Find the fourth.
4. The radius of a circle is 7 cm. What is its diameter?

12. Symmetry and Solids

Symmetry

- A **line of symmetry** divides a figure into two **mirror halves**.

Figure	Lines of symmetry
Equilateral triangle	3
Isosceles triangle	1
Scalene triangle	0
Square	4
Rectangle	2
Rhombus	2
Regular hexagon	6
Circle	Infinite

- **Letters with symmetry:** A, H, M, O, T, U, V, W, X, Y (vertical line); B, C, D, E, K (horizontal line)

Solids (3-D shapes)

Solid	Faces	Edges	Vertices
Cube	6	12	8
Cuboid	6	12	8
Triangular prism	5	9	6
Square pyramid	5	8	5
Triangular pyramid (tetrahedron)	4	6	4

- **Euler's formula: F + V − E = 2.** For a cube: 6 + 8 − 12 = 2 ✔️
- A cylinder, cone and sphere have **curved** surfaces.

> 🧠 **Memory Trick**
> **Euler: "Faces + Vertices = Edges + 2."** Remember it as "**F**unny **V**ideos **E**veryday = 2 hours".
> **Regular polygon: lines of symmetry = number of sides.**

> 🎲 **Play It in Class: Mirror Drawing**
> Fold a paper in half, draw half of a butterfly or rangoli along the fold, and cut it out. Open it to see perfect symmetry. Then: which letters of your name are symmetric?

✏️ Quick Check

1. How many lines of symmetry does a regular pentagon have?
2. A solid has 6 faces and 8 vertices. How many edges does it have?
3. Name two letters with no line of symmetry.
4. How many faces does a triangular prism have?

13. Mensuration: Perimeter and Area

Perimeter (the boundary length, in cm, m, ...)

Shape	Perimeter
Rectangle	2 × (length + breadth)
Square	4 × side
Triangle	Sum of the three sides
Equilateral triangle	3 × side
Regular polygon	Number of sides × side

Area (the space inside, in cm^2, m^2, ...)

Shape	Area
Rectangle	length × breadth
Square	side × side
Right-angled triangle	½ × base × height

- **Finding the missing side:** length = Area ÷ breadth, and side of a square = √Area
- **Units:** 1 m = 100 cm, but **1 m^2 = 10,000 cm^2**
- **Cost problems:** cost = area (or perimeter) × rate
- **Path around a rectangle:** area of the path = (outer area) − (inner area)

Example: a room is 6 m × 4 m. Perimeter = 2(6 + 4) = 20 m (for a border). Area = 6 × 4 = 24 m^2 (for tiles). Tiling at ₹150 per m^2 = 24 × 150 = **₹3,600**

🧠 **Memory Trick**

Perimeter = Fence (you walk around it). Area = Carpet (you cover it). Fencing, border, lace, wire → perimeter. Tiles, paint, carpet, grass → area.

📍 **Mumbai Connection**

A Mumbai 1BHK hall is 4 m × 3 m. How much does flooring cost at ₹200 per m^2? Area = 12 m^2, so cost = **₹2,400**.

🎲 **Play It in Class: Measure the Classroom**

Teams measure the classroom door, a desk and the blackboard with a measuring tape, then calculate the perimeter and area of each. The closest answer to the teacher's measurement wins.

✏️ Quick Check

1. Find the perimeter and area of a rectangle 15 cm × 8 cm.
2. The area of a square is 81 m^2. Find its side and perimeter.
3. The area of a rectangle is 96 cm^2 and its length is 12 cm. Find its breadth.
4. Find the cost of fencing a square park of side 50 m at ₹20 per metre.

14. Data Handling, Mean and Median

Collecting and organising data

- **Raw data:** data not yet arranged. An **array** is data arranged in ascending or descending order.
- **Tally marks:** |||| with a line across = 5, which makes counting easy
- A **frequency** table shows how many times each value occurs.

Showing data

- **Pictograph:** uses pictures or symbols. Always give a **key** (🍎 = 10 apples).
- **Bar graph:** bars of **equal width** with **equal gaps**. The height shows the value. Give it a title, label the axes and choose a scale.

Mean and median

- **Mean (average)** = sum of all the observations ÷ number of observations Mean of 4, 8, 6, 10, 12 = 40 ÷ 5 = **8**
- **Median** = the middle value after arranging the data in order
  - **Odd number of values:** the middle one. For 3, 7, 9, 12, 15 → **9**
  - **Even number of values:** the average of the two middle ones. For 2, 4, 6, 8 → (4 + 6) ÷ 2 = **5**
- **Range** = highest − lowest

🧠 **Memory Trick**

Mean = "fair share": put everything together and share it equally. **Median = "the middle of the road"**: line them up and stand in the middle. (Remember to **arrange first!**)

🎲 **Play It in Class: Class Survey**

Survey the class: "How many minutes do you take to reach school?" Make a tally table, draw a bar graph on chart paper, then find the mean and median. Is anyone a big outlier?

✏️ **Quick Check**

1. Find the mean of 12, 15, 18, 20, 25.
2. Find the median of 7, 3, 9, 1, 5.
3. Find the median of 10, 4, 8, 6.
4. In a pictograph 📘 = 20 books. How many books do 4½ symbols show?

Full-Book Revision Test

Section A: Quick answers (1 mark each)

1. Write the smallest 6-digit number in words (Indian system).
2. The product of two integers is −48. If one is −6, what is the other?
3. Which is the only even prime number?
4. Write 0.6 as a percent.
5. Find the supplement of 75°.
6. How many lines of symmetry does a square have?

Section B: Short answers (2–3 marks each)

7. Find the HCF and LCM of 18, 24 and 30.
8. Simplify: 3/4 − 2/3 + 5/6
9. Divide ₹2,700 in the ratio 4 : 5.
10. Solve: 3(x − 2) = 2x + 5
11. The angles of a triangle are in the ratio 2 : 3 : 4. Find the angles.
12. A rectangular garden is 25 m × 18 m. Find the cost of fencing it at ₹12 per metre.

Section C: Long answers (4 marks each)

13. Two bells ring at intervals of 15 and 20 minutes. They ring together at 8:00 am. When will they ring together next?
14. A train covers 360 km in 4 hours. Find its speed in km/h and in m/s.
15. The marks of 7 students are 45, 38, 50, 42, 47, 40, 46. Find the mean and median.

Answer Key

Ch 1: (1) Seven crore eight lakh forty-five thousand (2) 4,567,890 (3) 57,000 (4) XCIV

Ch 2: (1) 99,999 (2) (−12) + 7 + 5 = 0 (3) 24; −4 (4) −8, −1, 0, 3, 5

Ch 3: (1) By 4: yes (48 ÷ 4 = 12). By 9: yes (5 + 1 + 4 + 8 = 18). By 11: yes ((8 + 1) − (4 + 5) = 0). (2) 2 × 2 × 3 × 7 (3) HCF = 12, LCM = 72 (4) 4 × 48 ÷ 16 = 12

Ch 4: (1) {E, L, P, H, A, N, T}, n = 7 (2) No. It has one element, 0. (3) {x : x is the square of a natural number, x ≤ 25} (4) Equivalent (same count), not equal

Ch 5: (1) 45 : 120 = 3 : 8 (2) Riya ₹600, Aarav ₹800 (3) 4 × 15 = 60 and 6 × 10 = 60, so yes (4) 1 notebook = ₹30, so 13 notebooks = ₹390

Ch 6: (1) 4 5/6 (2) 3/4 (3) 7/3 + 7/4 = 28/12 + 21/12 = 49/12 = 4 1/12 (4) 1/2 + 1/2 = 1

Ch 7: (1) 1/8 (2) 17.03 (3) 70% (4) 15 m/s

Ch 8: (1) −7y (2) 10a + 2b (3) 2x − 4 (4) 2(9) − 3(−2) = 18 + 6 = 24

Ch 9: (1) x = 6 (2) 3x = 15, so x = 5 (3) x − 2 = 15, so x = 17 (4) 27

Ch 10: (1) 55° (2) 52° (3) x + 3x = 180, so 45° and 135° (4) Reflex angle

Ch 11: (1) 70° (2) (8 − 2) × 180° = 1080° (3) 360 − 285 = 75° (4) 14 cm

Ch 12: (1) 5 (2) F + V − E = 2, so 6 + 8 − E = 2 and E = 12 (3) F, G, J, L, N, P, Q, R, S, Z (any two) (4) 5

Ch 13: (1) Perimeter 46 cm, area 120 cm^2 (2) Side 9 m, perimeter 36 m (3) 8 cm (4) Perimeter 200 m, so cost = ₹4,000

Ch 14: (1) 90 ÷ 5 = 18 (2) In order: 1, 3, 5, 7, 9, so the median is 5 (3) In order: 4, 6, 8, 10, so (6 + 8) ÷ 2 = 7 (4) 90 books

Revision test: (1) One lakh (2) 8 (3) 2 (4) 60% (5) 105° (6) 4 (7) HCF 6, LCM 360 (8) 9/12 − 8/12 + 10/12 = 11/12 (9) ₹1,200 and ₹1,500 (10) 3x − 6 = 2x + 5, so x = 11 (11) 40°, 60°, 80° (12) Perimeter 86 m, so cost = ₹1,032 (13) LCM(15, 20) = 60 minutes, so at 9:00 am (14) 90 km/h = 25 m/s (15) Mean = 308 ÷ 7 = 44. In order: 38, 40, 42, 45, 46, 47, 50, so the median is 45